MASTER 2 - ADEO | CY TECH 2026

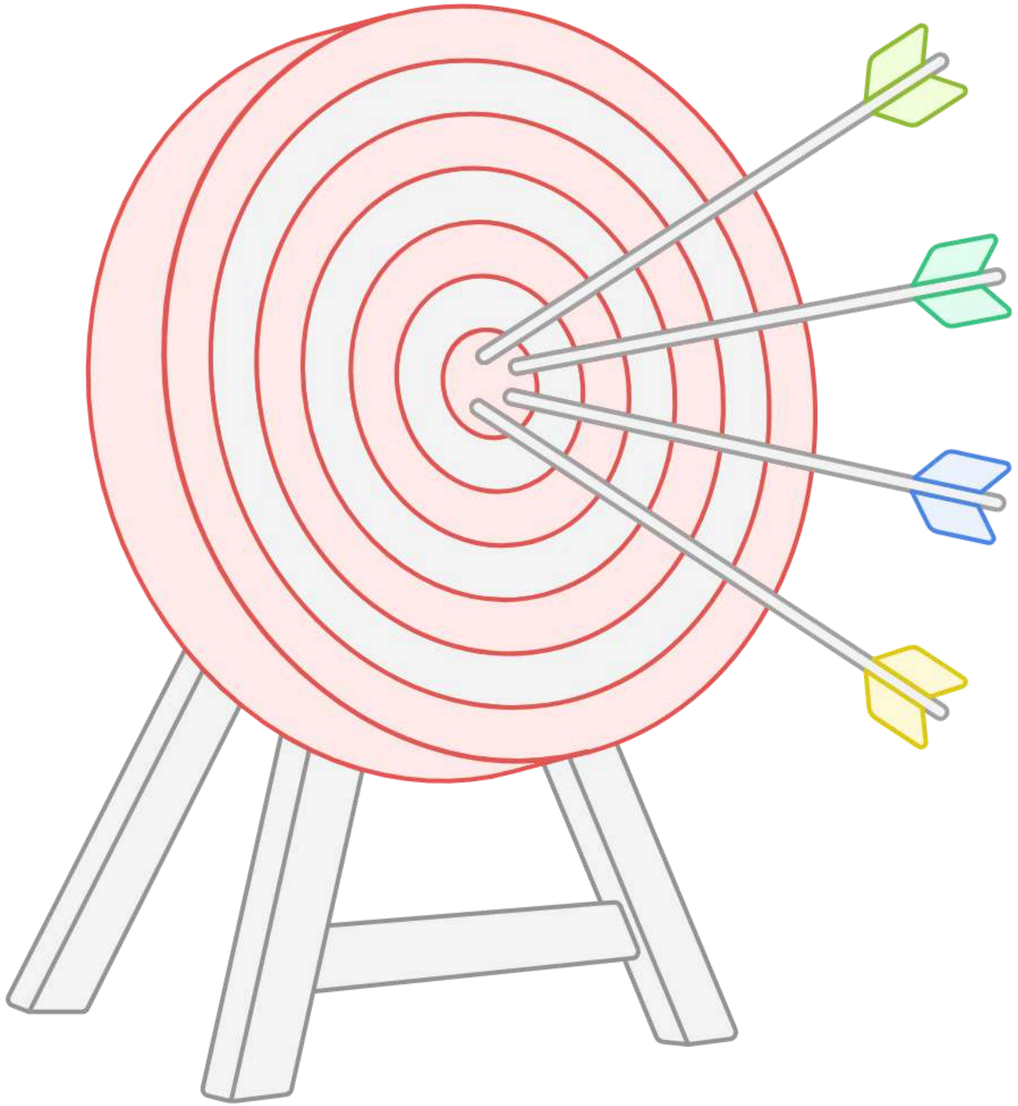
EUROPEAN AIR TRAFFIC ANALYSIS (1993-2022)

Project Presentation

PROFESSOR:
Madan Radhahkrishnan

PRESENTED BY:
Ahmed Farouk Damergi

Project Objective



Analyze European Air Traffic

Understand overall air traffic trends

Compare Passenger and Flight Activity

Examine the relationship between passenger numbers and flight operations

Identify Leading Countries and Trends

Pinpoint key players and emerging patterns

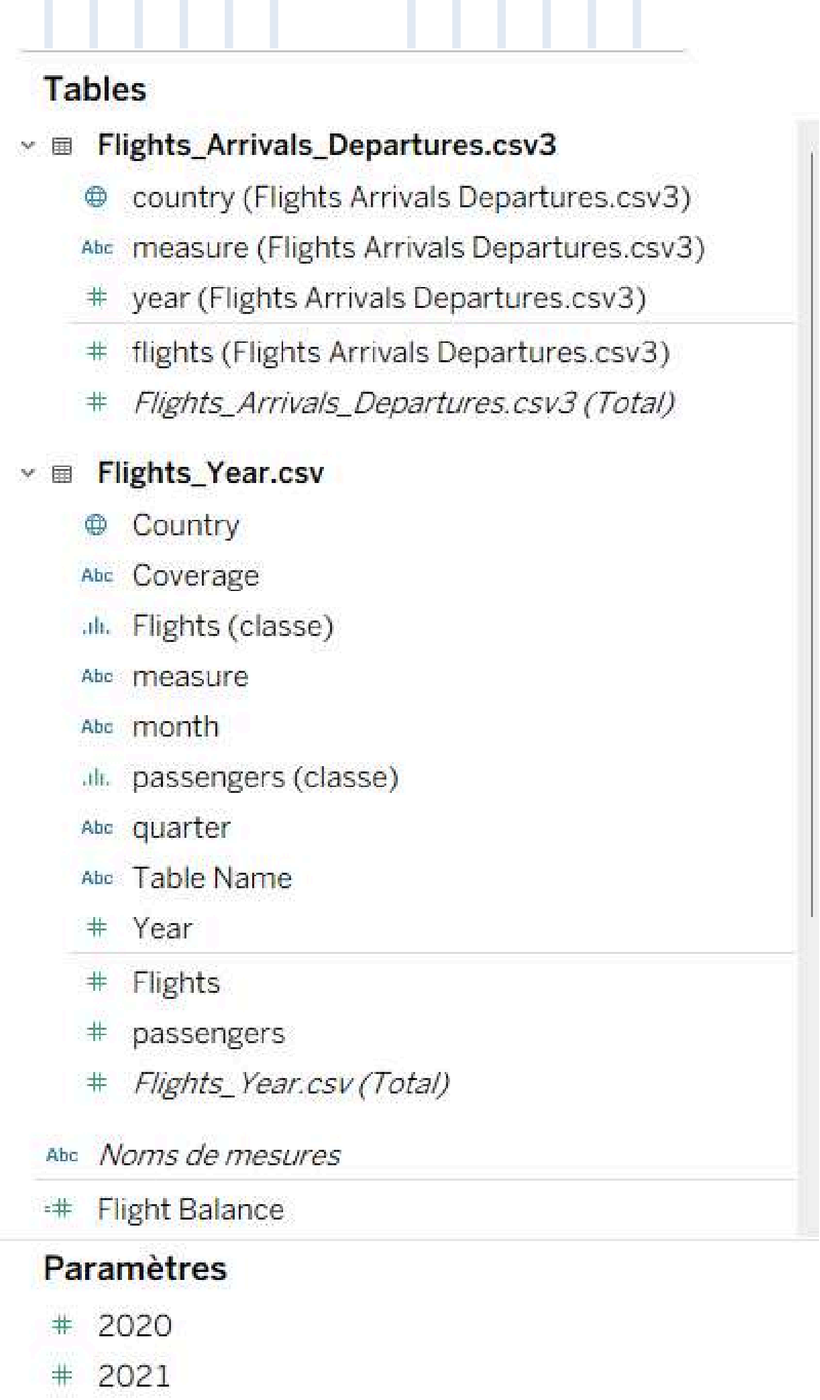
Detect Seasonality and Growth Patterns

Uncover cyclical trends and long-term growth

Dataset Description

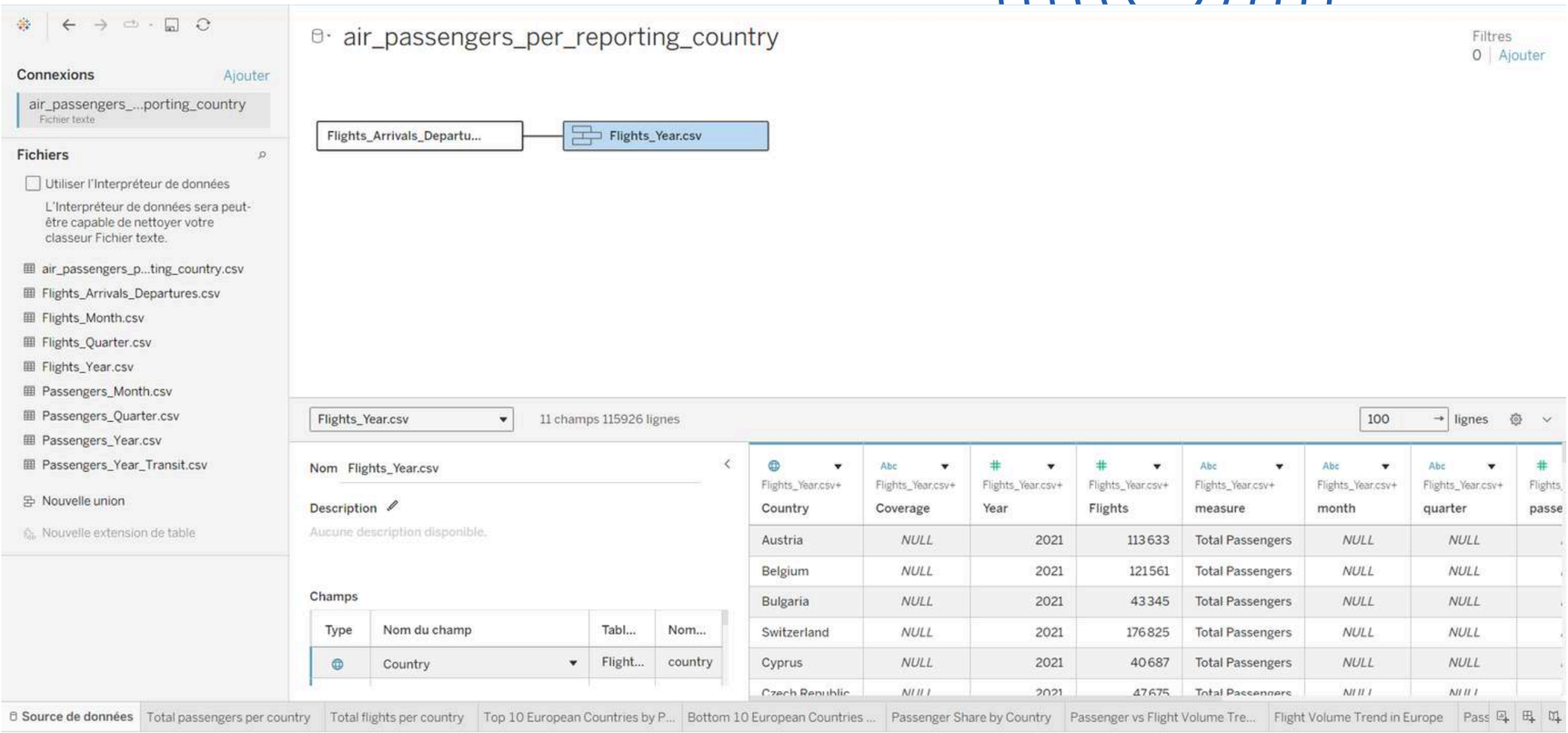
-Source: Aviation traffic dataset (European air traffic statistics)
-Time Period: 1993 - 2022
-Structure: Multiple CSV files (passengers, flights, arrivals, departures)
-Data Volume: Large dataset combining monthly, quarterly, and yearly reports
-Key Variables:

- Country
- Year / Quarter / Month
- Passengers
- Flights
- Arrivals and Departures



Data Preparation

- Imported multiple CSV datasets into Tableau
- Combined datasets using Union to integrate different time reports
- Verified data consistency across passengers and flights datasets
- Prepared variables for analysis (Country, Year, Passengers, Flights)
- Built a unified dataset used for all dashboards and KPIs
- Cleaned and validated the data before analysis



Calculated Measures

We created several calculated measures in Tableau to enrich the analysis:

- Flight Balance
- Passenger Growth
- Passenger Growth Index
- Passenger Market Share (%)
- Passengers per Flight

Flight Balance

```
SUM(IF [measure] = "Arrivals" THEN [Flights] END)
-
SUM(IF [measure] = "Departures" THEN [Flights] END)
```

Passenger Growth

Les résultats sont calculés avec Table (horizontale).

```
SUM([passengers]) - LOOKUP(SUM([passengers]), -1)
```

Passenger Growth Index

Les résultats sont calculés avec Table (horizontale).

```
SUM(passengers) /
WINDOW_MIN(SUM(passengers)) * 100
```

Passenger Market Share

Les totaux résument les valeurs de Table (horizontale).

```
SUM([passengers]) / TOTAL(SUM([passengers]))
```

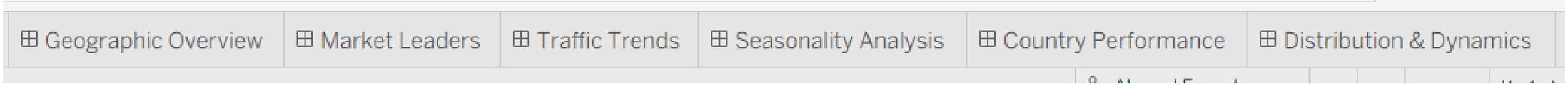
Passengers per Flight

```
SUM([passengers]) / SUM([Flights])
```

Dashboard Structure

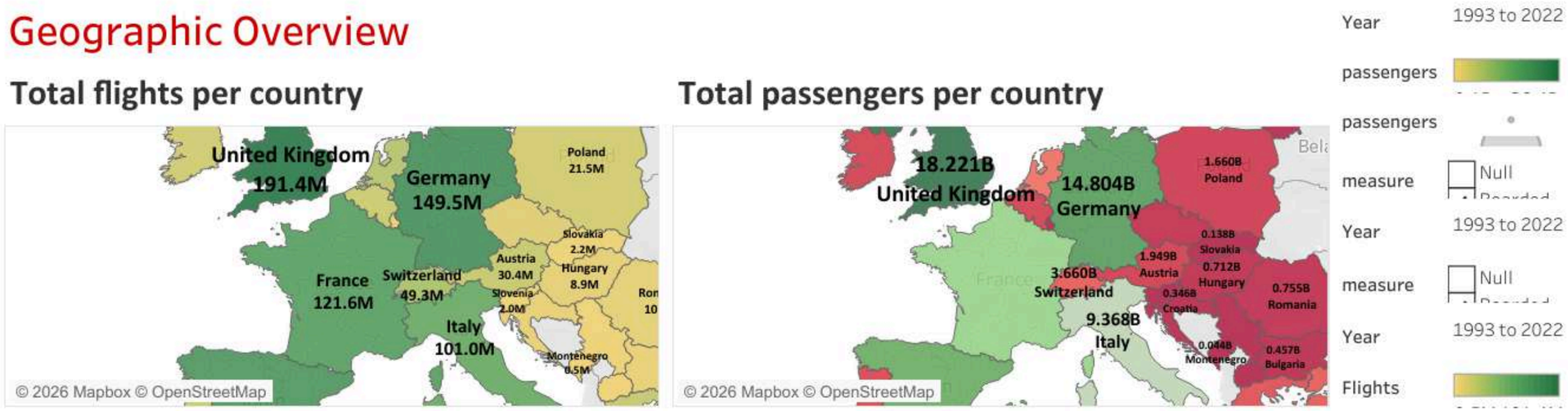
To analyze the data efficiently, we organized the visualizations into six dashboards, each focusing on a specific aspect of air traffic analysis:

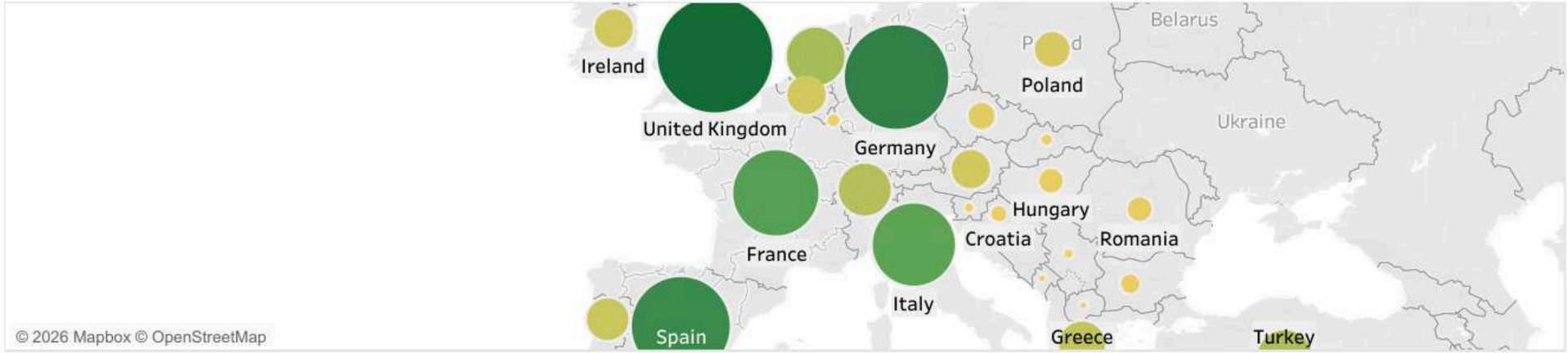
- Geographic Overview : spatial distribution of passengers and flights
- Market Leaders : comparison of top and bottom countries
- Traffic Trends : evolution of passengers and flights over time
- Seasonality & Growth : seasonal patterns and growth indicators
- Country Performance : efficiency and country comparisons
- Distribution & Dynamics : traffic distribution and country evolution



Geographic Distribution of Air Traffic

- Air traffic distribution across European countries
- Identification of major aviation hubs
- Comparison of passenger and flight volumes geographically



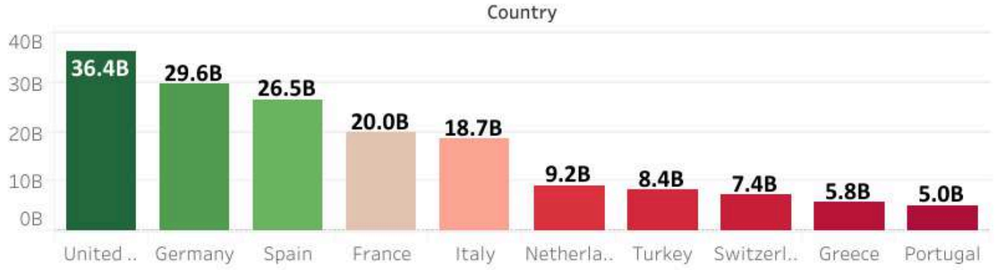


Market Leaders in European Air Traffic

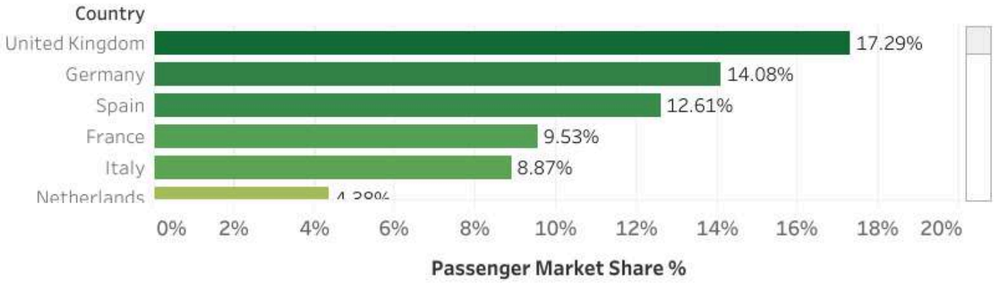
- Passenger traffic concentrated in a few countries
- United Kingdom leads European passenger volume
- Large gap between top and bottom countries

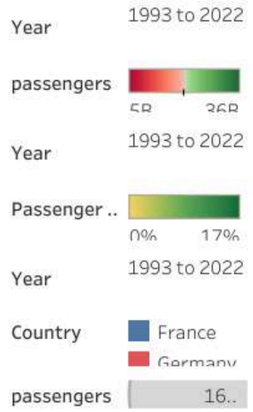
Market Leaders

Top 10 European Countries by Passenger Volume

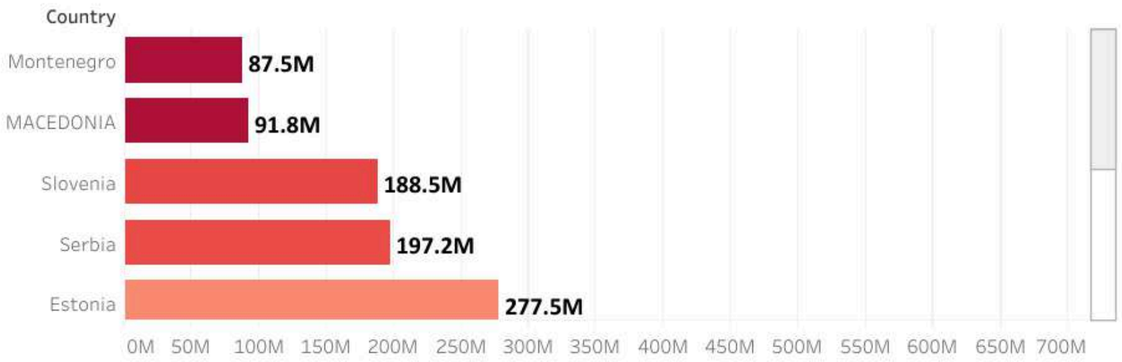


Passenger Market Share by Country (%)

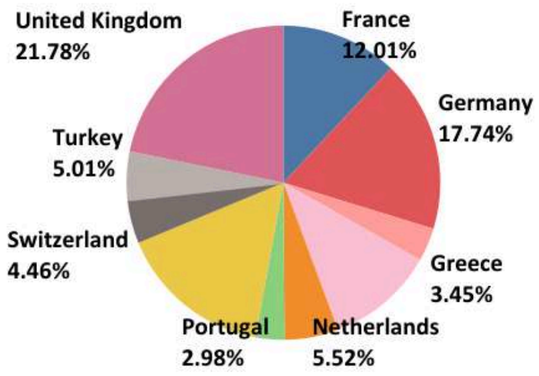




Bottom 10 European Countries by Passenger Volume



Passenger Share by Country

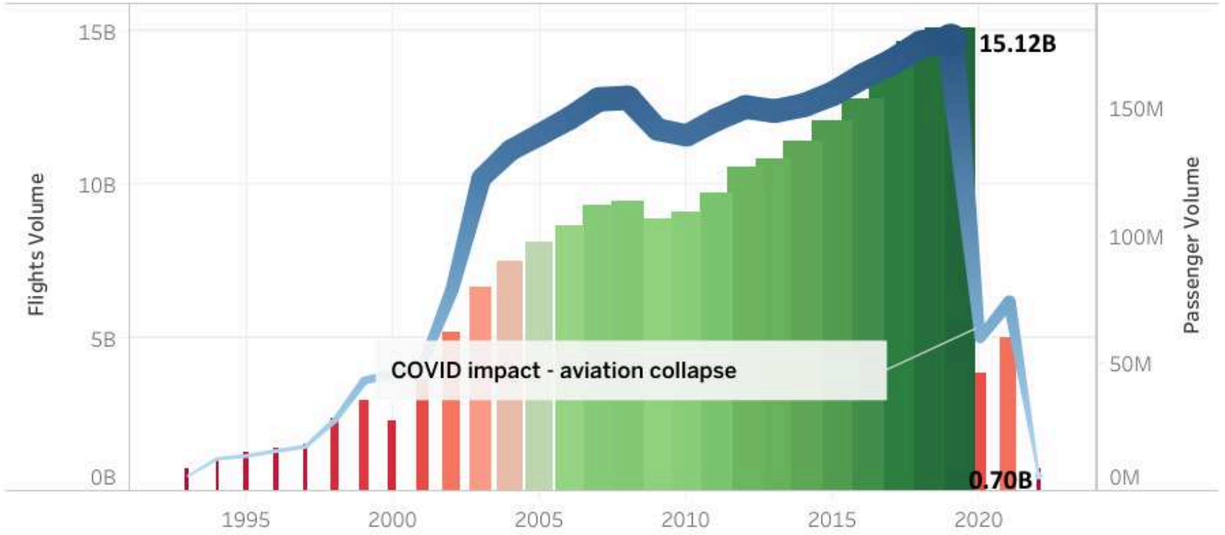


Air Traffic Trends Over Time

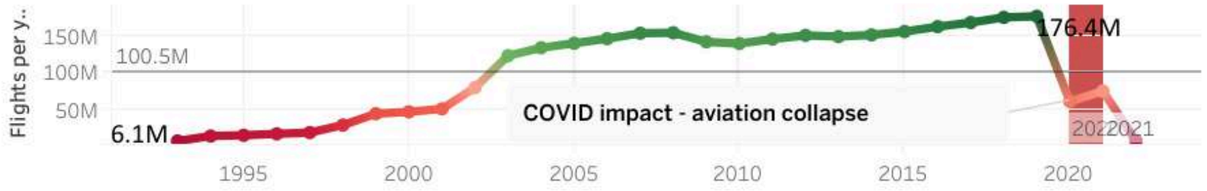
- Air traffic increased steadily until 2019
- Sharp decline during COVID-19 period
- Gradual recovery after 2021

Traffic Trends

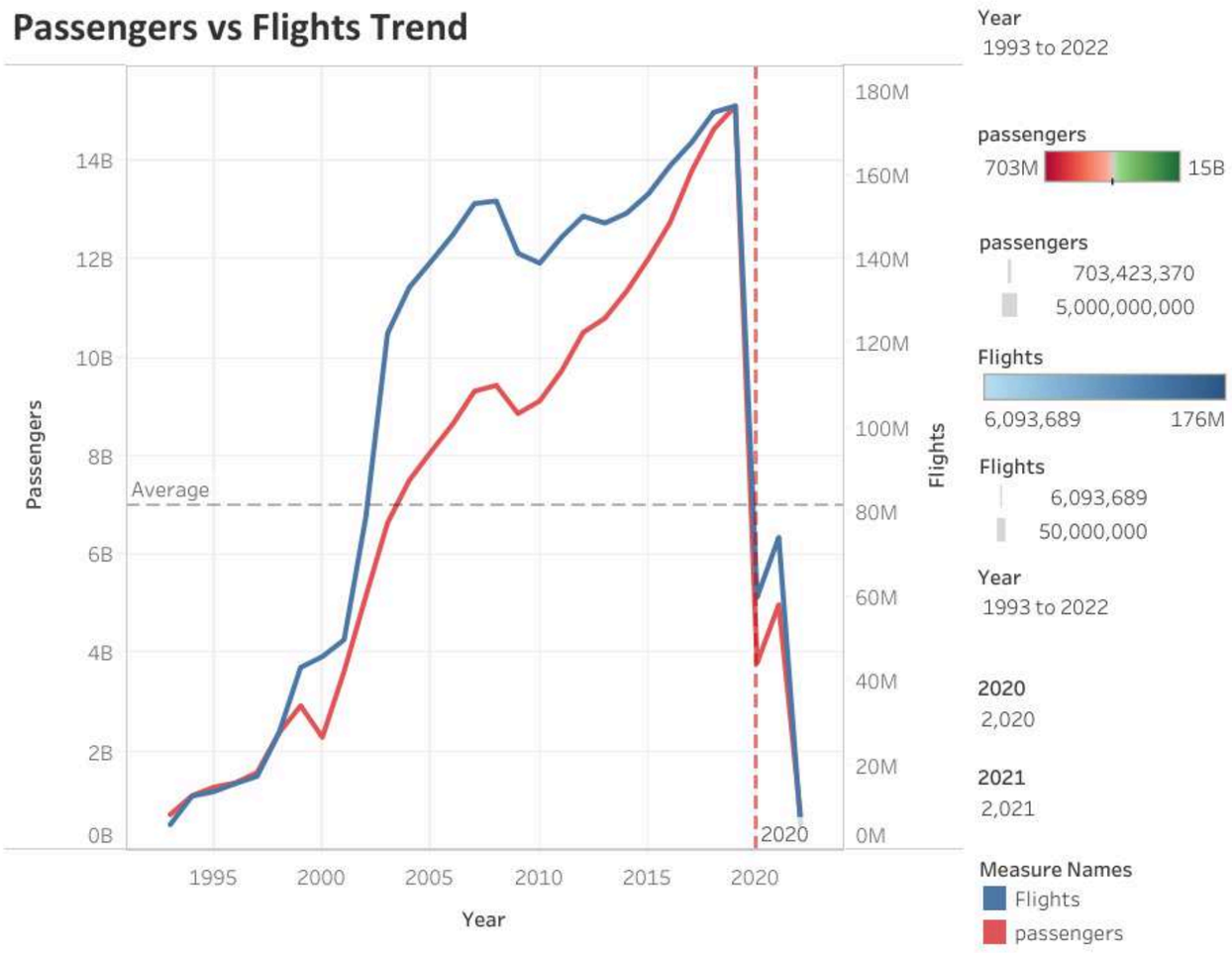
Passenger vs Flight Volume Trend Over Time



Flight Volume Trend in Europe



Passengers vs Flights Trend

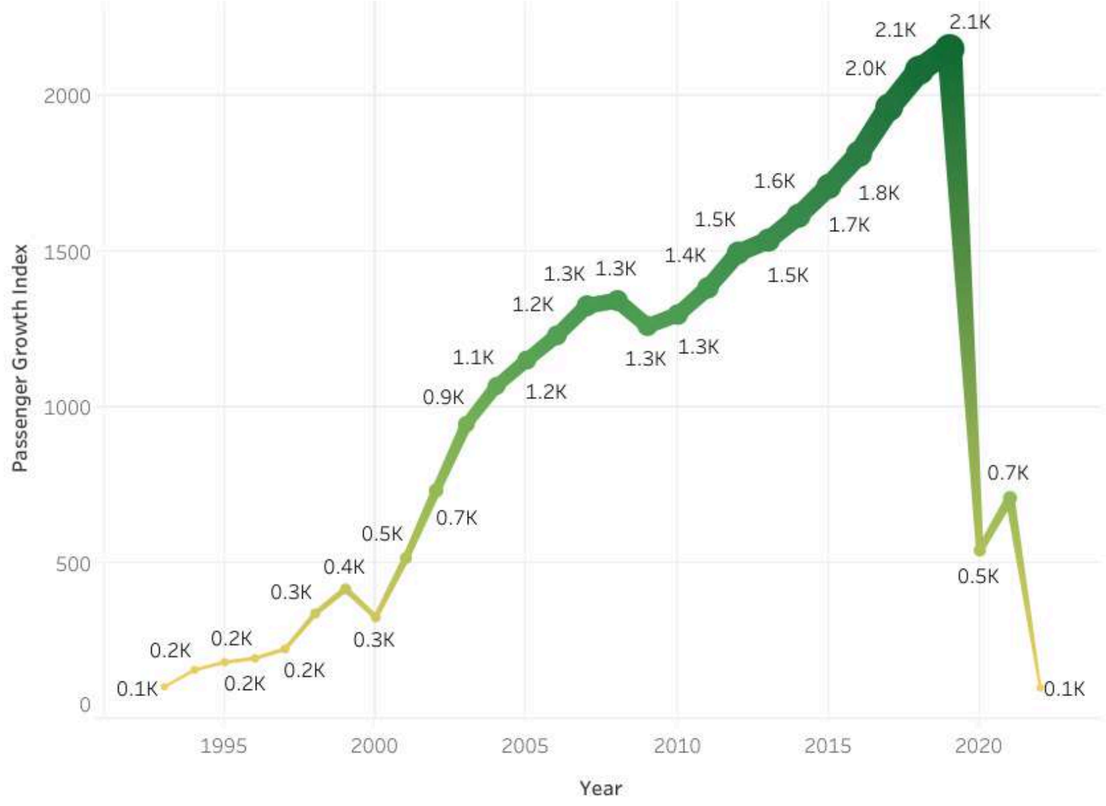


Seasonality and Growth Patterns

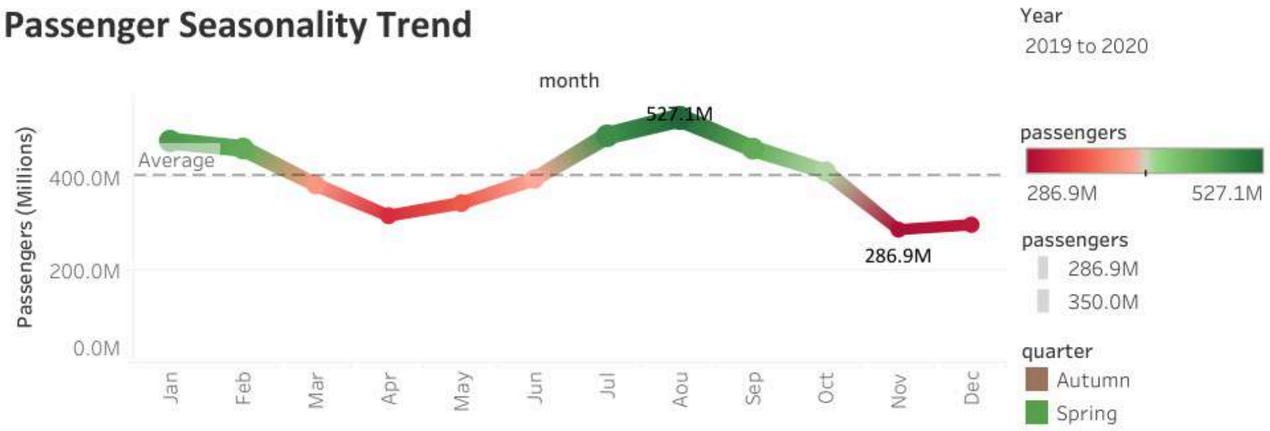
- Strong seasonal peaks during summer months
- Lower traffic during winter periods
- Long-term growth trend visible before 2020

Seasonality Analysis

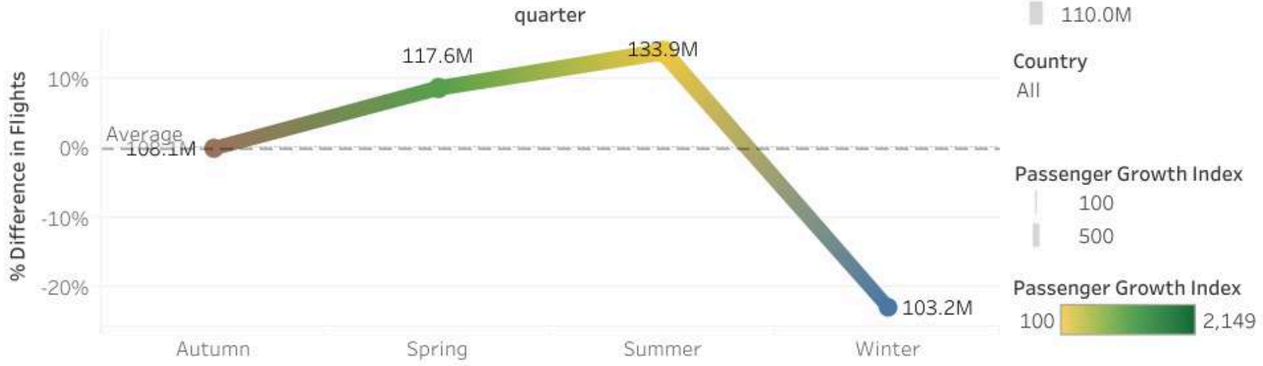
Passenger Growth Index



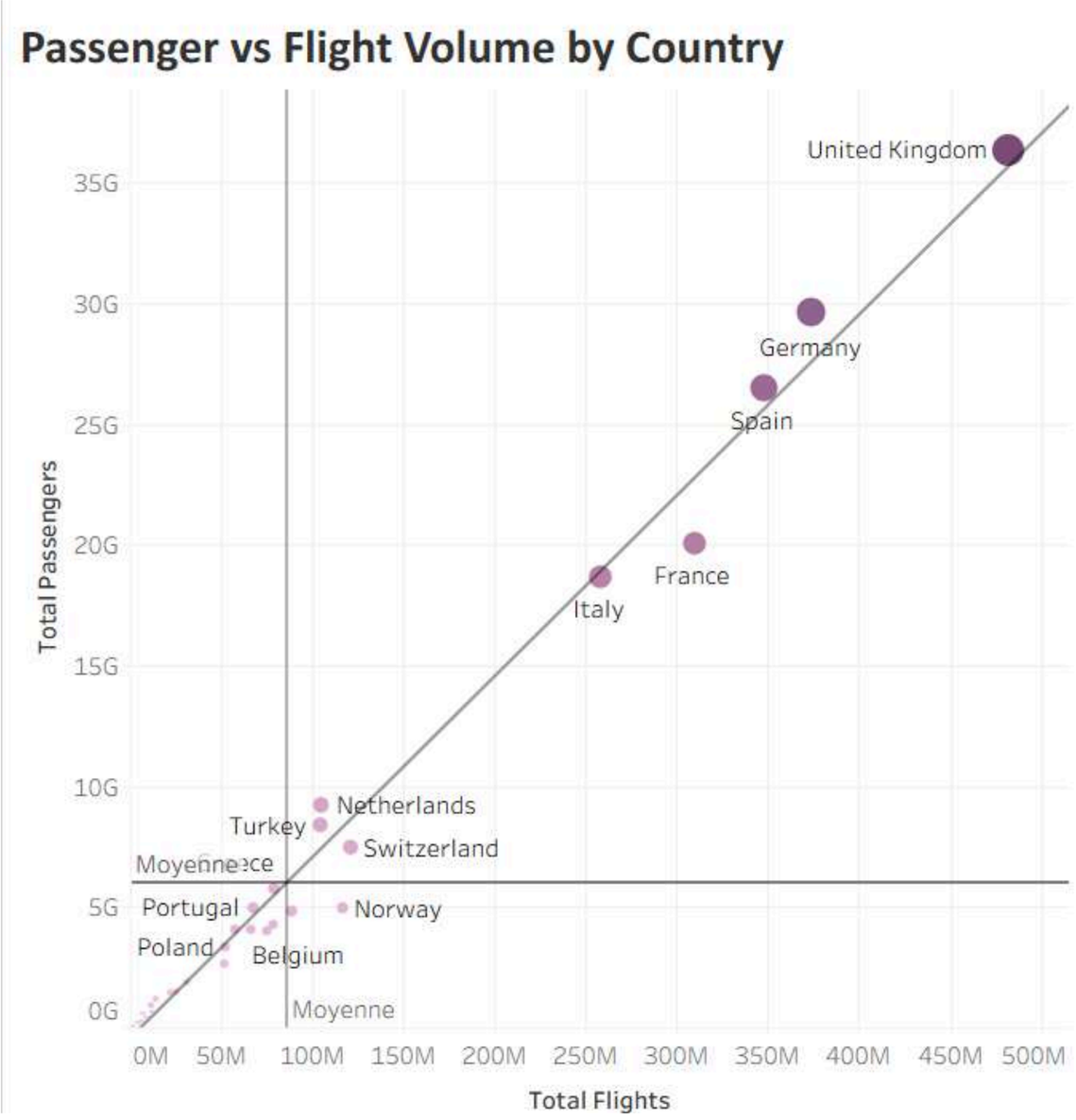
Passenger Seasonality Trend

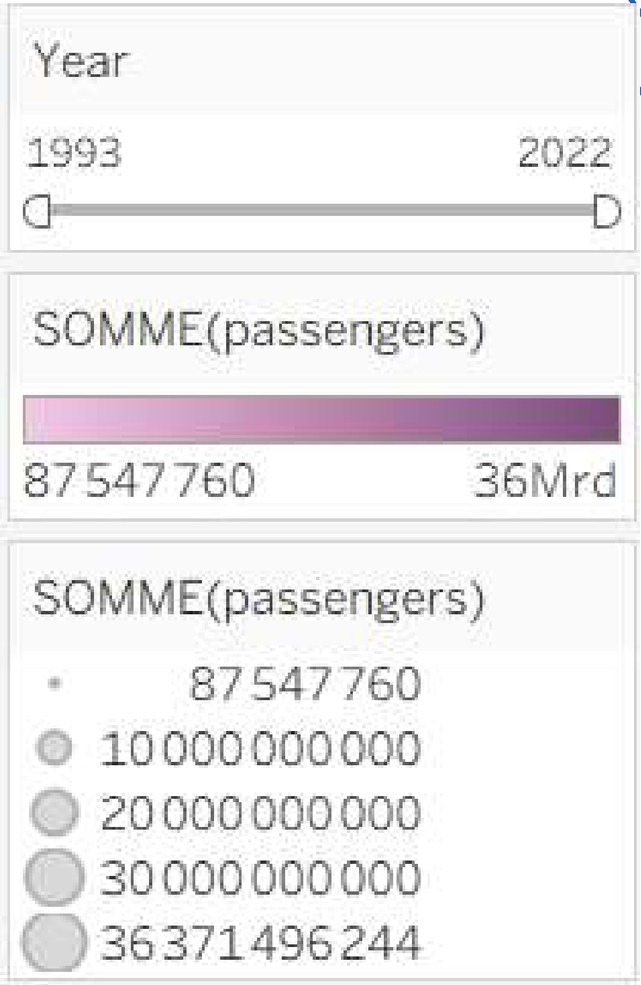


Quarterly Flight Seasonality



Country Performance





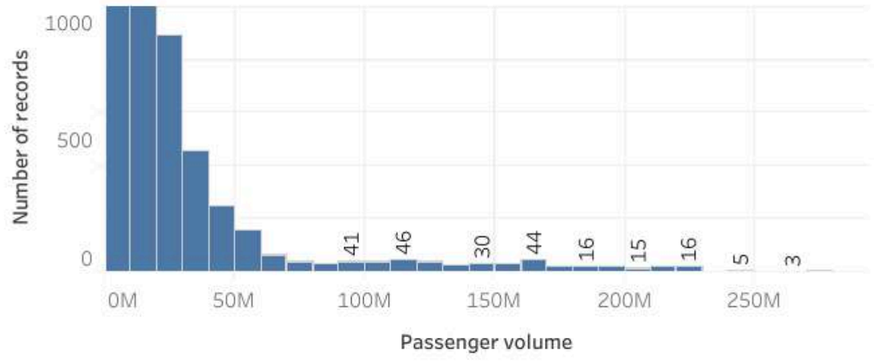
- Strong correlation between flights and passenger volume
- Efficiency varies across countries
- Some countries transport more passengers per flight

Traffic Distribution

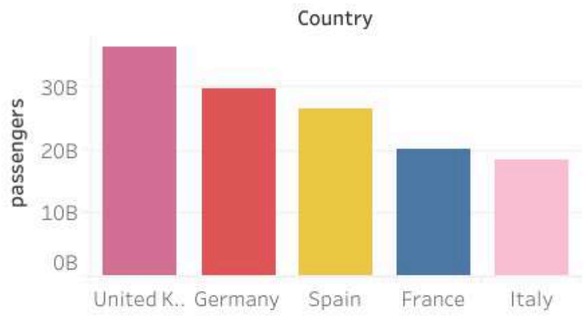
- Passenger traffic highly concentrated among a few countries
- Long-term growth visible until 2019
- Significant drop during COVID-19 period
- Gradual recovery observed after 2021

Distribution & Dynamics

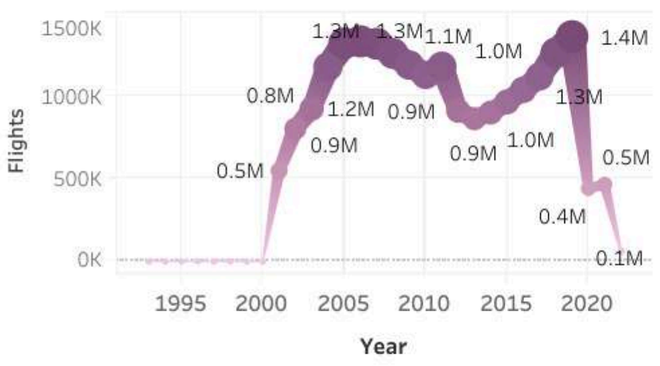
Passenger Volume Distribution



Top 5 Passenger Countries per Year



Flights Evolution by Country



Year
2010 to 2022

passengers
225,470 2B

Country
United Kingdom
Germany

Country
Austria
Belgium

Passenger Volume Heatmap by Country and Year

Passenger Evolution by Country

passengers
0.0B
0.5B

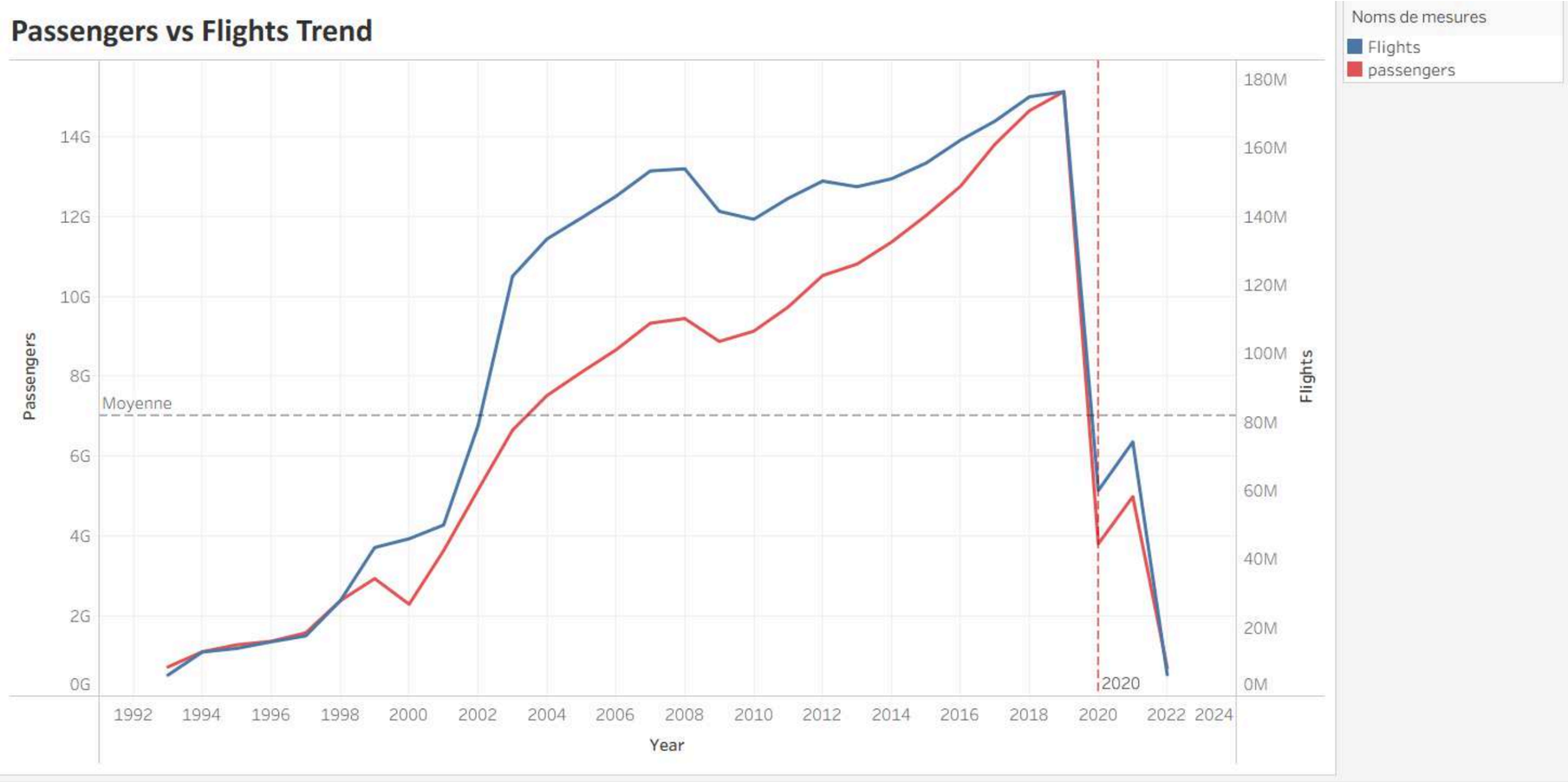
Country
Austria
Belgium

Flights
61,140 1,356,711

Flights
61,140
500,000

Key Insights

- European air traffic is concentrated in a few major countries
- United Kingdom, Germany and Spain dominate passenger traffic
- Strong long-term growth from 1993 to 2019
- Sharp collapse in 2020 due to COVID-19
- Recovery trend after 2021

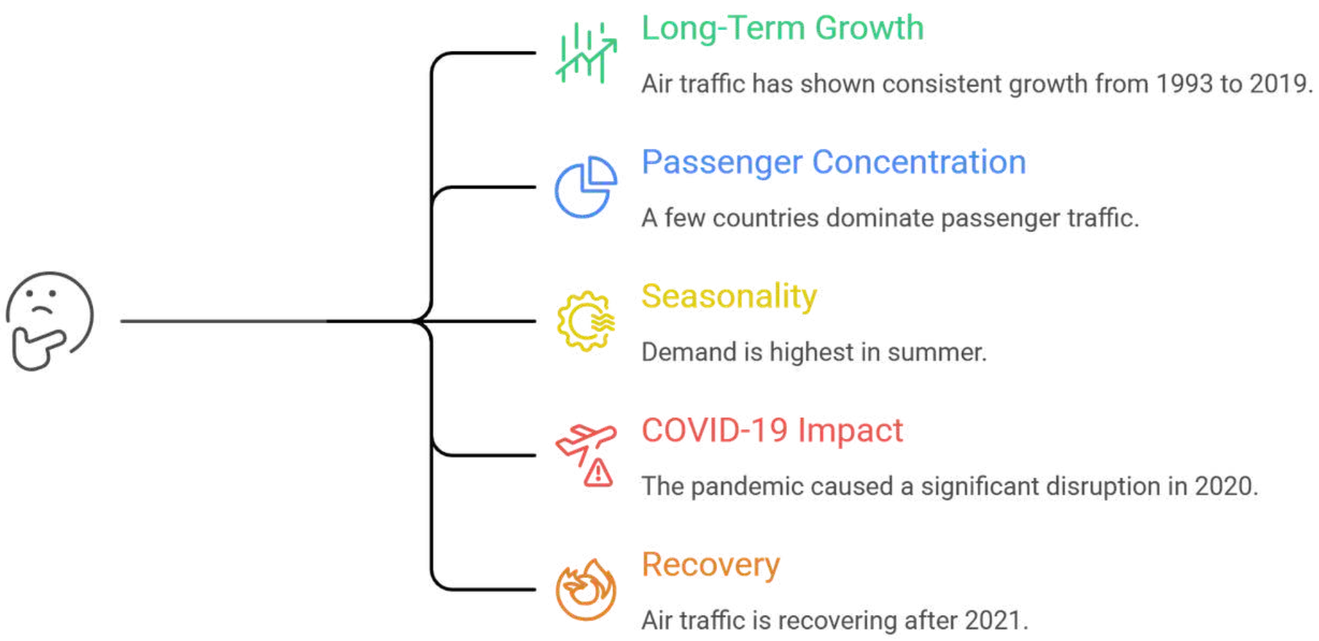


Additional Insights

- Clear seasonality pattern in European air traffic
- Passenger traffic peaks during summer months
- Winter months show lower demand
- Strong correlation between flights and passenger volume

Conclusion

- European air traffic shows strong long-term growth from 1993 to 2019
- Passenger traffic is highly concentrated among a few countries
- Seasonality strongly influences demand, especially in summer
- COVID-19 caused a major disruption in 2020
- Air traffic shows signs of recovery after 2021



Thank you